Linear Algebra MAT 226

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Exercise 2.7

Q4) translate by $(-1, 4)$ the 3×3 matrix T is

$$T = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

scale x coordinate by $1/2$ and y coordinates by $3/2$

$$S = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$ST = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 3/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/2 & 0 & -1/2 \\ 0 & 3/2 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

Exercise 3.1

Q2) cofactor expansion across the 1st row

$$\begin{bmatrix} 0 & 5 & 1 \\ 4 & -3 & 0 \\ 2 & 4 & 1 \end{bmatrix} = 0\begin{bmatrix} -3 & 0 \\ 4 & 1 \end{bmatrix} - 5\begin{bmatrix} 4 & 0 \\ 2 & 1 \end{bmatrix} + 1\begin{bmatrix} 4 & -3 \\ 2 & 4 \end{bmatrix}$$

$$= 0(-3+0) - 5(4-0) + 1(16+6)$$

$$= 0(-3) - 5(4) + 1(22)$$

$$= -20 + 22$$

$$= 2$$

cofactor expansion down the second column.

$$\begin{bmatrix} 0 & 5 & 1 \\ 4 & -3 & 0 \\ 2 & 4 & 1 \end{bmatrix} = 5\begin{bmatrix} 4 & 0 \\ 2 & 1 \end{bmatrix} - 3\begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} - 4\begin{bmatrix} 0 & 1 \\ 4 & 0 \end{bmatrix}$$

$$= -5(4-0) - 3(0-2) - 4(0-4)$$

$$= -5(4) - 3(-2) - 4(-4)$$

$$= -20 + 6 + 16$$

$$= +2$$

Q40) (a) False

According to cofactor expansion theorem, the determinant of an $n \times n$ matrix

can be computed using a cofactor expansion of any row or column and expansions produce the same value.

(b) False

No, it is the product of entries in the main diagonal. (theorem 2).

Q42) consider the coordinates of the parallelogram's vertices:

$O = (0,0)$ $v = (c,0)$ $u = (a,b)$

$u+v = (a+c, b)$

because v lies on x axis, it can be the base of parallelogram.

length of base is c

height is y coordinate of u which is b.

since a, b, c are positive,

$$\text{area} = c \times b = bc$$

to compute the determinants

matrix $[u\ v]$

$$\text{determinant} \begin{bmatrix} a & c \\ b & 0 \end{bmatrix} = a \times 0 - b \times c = -bc$$

matrix $\begin{bmatrix} v & u \end{bmatrix}$

determinant $\begin{bmatrix} c & a \\ 0 & b \end{bmatrix} = c \times b - a \times 0 = bc$

